Engineering Mathematics - I · Unit_02 · Unit 2

DRAFT — UE26MA141A · Unit_02 · Unit 2 · Particular integral when X is any other form Continued

Standalone video lesson · authored study notes with source citations

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

Contents

1. 1 · Particular integral when X is any other form Continued
2. Appendix A · Symbol glossary
3. Appendix B · Sources, coverage and gaps
4. Appendix C · Revision and retrieval practice

M1

Particular integral when X is any other form Continued

1 Particular integral when X is any other form Continued

Study focus: Particular integral when X is any other form_Continued. This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment.

EV0000158 · EV0000159

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

In this session, we will take up problems of finding particular integral when the right hand side as capital X doesn't belong to any of the four standard forms we have discussed earlier.

Key details

- It means when x is not a power ax, when x is not sine of a x plus b as well as of x plus b.
- When x is not a polynomial case, when x is not e power x into v, where v is a function of x.
- any one of these four forms.
- D plus one times d plus two of y equals sine of v power x.
- This is not any one of these four forms.
- The particular integral is written as 1 divided by f of b which is nothing but d plus 1 times d plus 2 of capital X.
- So we first resolve this into partial fraction.
- Once we use partial fraction.
- This expression reduces to one divided by d plus one minus one divided by d plus two.
- It means a value becomes one, b becomes minus one.
- Therefore, the particular integral is written as 1 divided by d plus 1 minus 1 divided by d plus 2 of this capital X.
- Minus 1 by d plus 2 of sin of v power x.
- We can take this as p2.
- Finally, after computing p1 and p2, we go for p1 minus p2.
- We know a standard formula that is 1 over p minus 1 of f of x equals e power ax integral e power minus ax into f of x to d x.

- So when you apply this formula for this particular case we get a power minus x because area is minus 1.
- Therefore a power minus a x becomes e power minus x.
- A power minus ax becomes a power x, then sine of e power x times dx.
- So we need to integrate this by substitution method.
- Then this expression becomes integral psi t into dt.
- Therefore, the value of P1 is going to be minus a power minus x of cos of e power x.
- One over D plus two of sine of P power X.
- Therefore, as per the standard formula, we get this particular step.
- So when we use substitution method that is a power x equal to t, so that reduces to a power minus 2x integral.

Example or application from the lecture

- It means let us learn how to find out particular integral when the right hand side x is not.
- Let us compute 1 divided by p plus 1 of sine of p power x.
- Let us take e power x equal to t so that dt becomes e power dx.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000158 · EV0000159 · EV0000160 · +1 source units in pack.json

Core explanation · 05:00–09:59

t s t dt.

Key details

- Therefore, the final PI is going to be P1 minus P2.
- We get this as the answer.
- Provided we have negative sign outside.
- So therefore the particular integral for this problem is minus e power minus 2x into sine of e power x.
- Find the particular integral for the differential equation d square plus phi d plus six of y equals e power 2x times sin of e power x.
- Similar to last problem.
- Finally, Pi can be written as 1 by D plus 2 minus 1 by D plus 3 of E power 2x times sine of E power x.
- In a similar manner, for this case we get e power x times e power 4x times e power.
- E power x into dx.
- After using the same substitution, it reduces to integration of t power four sin t.
- Even this can be solved by integration by partnoli rule.
- The integration of first one is going to be this.
- When we use x t to be e power x, we get this particular step.
- To find out PI when the right hand side X doesn't belong to any of the four standard forms we have discussed earlier.

Example or application from the lecture

- So that leads to minus t cos t plus integral cos t dt because very well how to apply integration by parts let's use the technique so finally we get e power minus 2x of minus t cos t and integral of cos t becomes sine t so finally this is same as e power minus 2x times this.
- In a similar manner, let us solve one more one more problem to get a clarity.
- Let us find the factors of this d plus 3 times d plus 2 of y equals right hand side.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000162 · EV0000163 · EV0000164 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Particular integral when X is any other form_Continued transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_Particular_integral_when_X_is_any_other_form_Continued__5c24666421028690.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral when X is any other form_Continued transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_Particular_integral_when_X_is_any_other_form_Continued__5c24666421028690.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Particular integral when X is any other form_Continued transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_Particular_integral_when_X_is_any_other_form_Continued__5c24666421028690.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9967; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Particular integral when X is any other form_Continued transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_02/AV_Summaries/009_Particular_integral_when_X_is_any_other_form_Continued__5c24666421028690.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
2. 2 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
3. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
4. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
5. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
6. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00038	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00039	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare...
SRC00040	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare...
SRC00041	2 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0020 to s0036. Sidecar variants compare...

C Revision and retrieval practice